

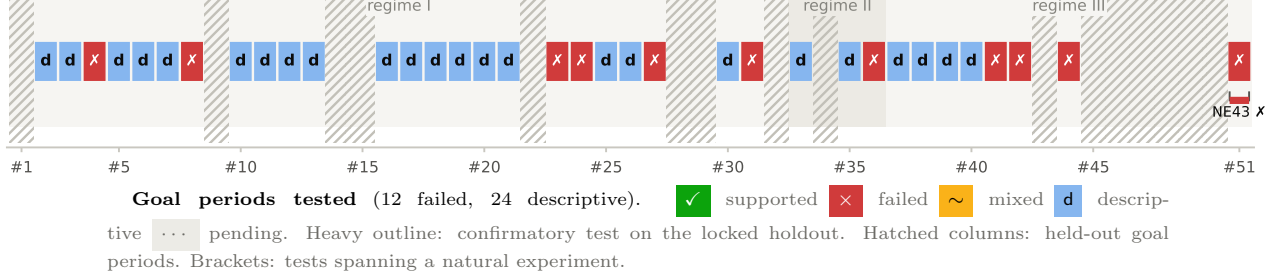
H43: Kicks leave a refractory window

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 2.5/5

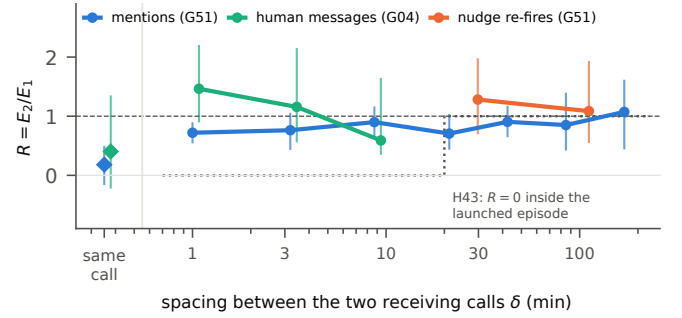
Bottom line. Kicks leave no refractory window: a second kick read by a later model call works about as well as the first, and only kicks absorbed into the same read are wasted.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Does a kicked agent ignore a second kick until the task the first one launched is over? Every kick was timed at the model call that first read it (DQ1 ledger). The second kick's effect, as a function of spacing δ , was compared with an isolated first kick under past-only matching. The synthetic validation (planted windows) came first. Replication covered 35 non-holdout periods, with native tests on NE43 (nuder off), G38 (pause gates) and G04 (human chat). No refractory window was found. A second kick read by a later call keeps 80–130% of the effect at every δ from 1 min to 4 h: mentions $R = 0.78$ [0.67, 0.89] (15 periods), human messages $R \approx 1.4$, nudge re-fires 1.28. Only a kick read in the *same* call adds little ($R = 0.2$ –0.4). The episode lock is absent (in-episode 0.67 vs 0.83). Holdout not run.



Observables. Second-kick effect relative to a first kick (lnHR ratio, 95% CI), against the spacing of the two receiving calls. Diamonds: the second kick read in the *same* call. Dotted: H43's prediction ($R = 0$ inside a 20-min launched episode). Mentions (G51) dip only to ~ 0.8 (pooled over 15 periods: 0.78); human messages (G04) and nudge re-fires (G51) stay at or above 1.

PHYSICAL MODEL

Model 09 (renewal / refractory Hawkes) with model 03's refractory compartment. A kick read at time t multiplies the agent's escape (or reply) hazard by a gain g_c with a short kernel κ . A recovery function r is measured from the previous effective kick. H43 claims $r = 0$ inside the launched task episode (length L). The rivals are a habituation clock $r(\delta)$, $r \equiv 1$ (state dependence only) and read-out batching (kicks read in one call count once). $R(\delta) = E_2/E_1$ uses pooled log hazard ratios.

$$\lambda_{\text{esc}}(t) = \lambda_0(a) [1 + g_c \kappa(t - t_{\text{read}}) r(t - t_{\text{last}})]$$

$$\text{H43: } r = \mathbb{I}[t - t_{\text{last}} > L]; \quad \text{habituation: } r = 1 - a e^{-\delta/\tau}$$

$$E = \ln \frac{\sum \text{events}_K / \sum \text{at-risk}_K}{\sum \text{events}_C / \sum \text{at-risk}_C}$$

$$R(\delta) = E_2(\delta)/E_1, \quad \delta_{1/2} : R(\delta_{1/2}) = \frac{1}{2}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Useful as a negative design rule. If you run a swarm, you do not need to wait for an agent to finish its task before nudging or messaging it again. A kick read by a later call works about as well as the first. What wastes a kick is sending it before the agent's next model call, e.g. during its PAUSE: it lands in the same read as the earlier one and adds little. Merge such messages or deliver them later. The nudger's 15-min floor is already past any refractoriness. Its real limit is that a first nudge buys a glance (escape $\times 1.75$), not sustained work. The physics label (refractory renewal) did not survive. What remains is the mundane fact that the context assembly is the unit of attention.

RESULTS VS PREDICTIONS

Prediction	Observed	
P1 window, $R_{\text{short}} \leq 0.5$	mentions 0.57–0.79; humans, nudges ≥ 1	×
P2 window $\approx$ episode	$\delta_{1/2} = 0$ vs $\tilde{L} = 5\text{--}56$ min	×
P3 episode lock	R_{in} 0.65–0.69 vs R_{act} 0.81–0.84	×
P4 same-call kick ≤ 0.3	0.18 (reply), 0.31 (talk)	✓
P5 synthetic recovery	nulls clean; mention windows only	~
P6 NE43 nudge window	no sustained E_1 ; re-fire $R = 1.28$	×
P7 invariance across NE43	mention $\Delta R = \pm 0.16$	✓
P8 G38 pause gates	8 re-kicked gates: untestable	n/a
P9 G04 human trains	$R(\delta \leq 2 \text{ min}) = 1.46$	×
P10 rule reliable	no window in 98–100% of draws	✓

GOAL PERIODS IN DETAIL

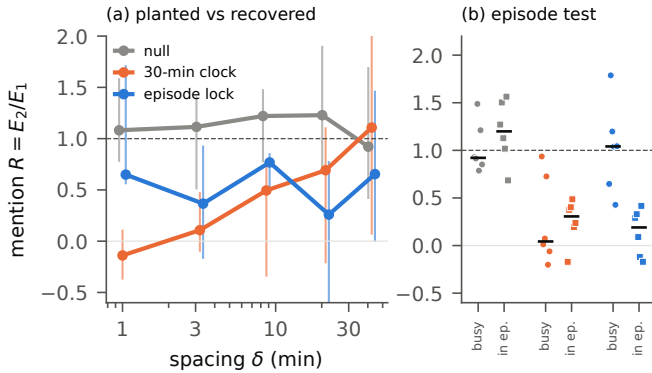
G02 d descriptive
G03 d descriptive
G04 × failed
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 × failed
G10 d descriptive
G11 d descriptive
G12 d descriptive
G13 d descriptive
G16 d descriptive
G17 d descriptive
G18 d descriptive

+22 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. Synthetic validation at G51 scale. (a) Mention $R(\delta)$: the null stays near 1. A planted 30-min habituation clock is recovered ($\delta_{1/2}$ 15 vs 18 min); an episode lock gives an intermediate curve. (b) The episode test separates the three worlds. The real data look like the null in both panels.

CAVEATS

Only mention curves are identifiable at G51 size. Human messages are testable in G04 alone (5-min quiet rule, amended before outcomes), nudges marginally. The nudge first-kick effect exists for “any activity” (post hoc) but not for sustained work (pre-registered), so R there rests on the post hoc outcome. Re-fired nudges outperforming first ones may reflect the nudger’s unobserved text trigger. Estimated call starts for non-Gemini agents misclassify some same-call vs next-call pairs, pulling $R(\delta \leq 2)$ toward the batched value. Episodes come from minute-grid runs, not task content. 35 periods $\times$ 3 classes $\times$ 5 outcomes: replication verdicts are templated and not independent.

ROUND 2

Price the k -th kick inside one read (per-provider timing). Test nudge facilitation against the trigger (H35). Run content-defined episodes and a content-drift outcome. Run `confirm.py`.

- **Essence.** the unit of operator attention. Round 1 says it is the recipient’s read-out (one model call), not its task episode; round 2 should confirm that on the holdout and price the k -th kick inside one read.
- **H43-R1** Run the confirmatory script.
- **H43-R2** Nudge facilitation or nudger selection?
- **H43-R3** Read-out batching as the object.
- **H43-R4** Content-defined episodes and a content outcome.
- **H43-R5** Pooled gate test.
- **H43-R6** Decompose the NE43 swarm drop.